NNABUIKE CHRISTIAN NGWU

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EDUCATION

University of Nigeria, Nsukka (UNN), Enugu State, Nigeria **Jan. 2026 to Present**
Master of Engineering (M. Eng.) in Mechanical Engineering
Expected completion: May 2027
Field: Thermofluids and Energy Engineering
Thesis: Comparative Analysis of Battery and PCM Energy Storage in PV-Powered Cold Rooms for Sustainable Microgrids
Advisor: Professor Howard Njoku

University of Nigeria, Nsukka (UNN), Enugu State, Nigeria **Sept. 2018 to Mar. 2024**
Bachelor of Engineering (B. Eng.) in Mechanical Engineering
Final Grade: First Class Honours **(CGPA: 4.69/5.0)** **Class Rank: (1st/119)**
Thesis: Analysis of Potentials for Maintaining Indoor Comfort in Green Buildings
Advisor: Professor Howard Njoku

RESEARCH INTERESTS

Energy Modelling | Energy in Buildings | Clean Energy | Renewable Energy Systems | Energy Sustainability | Energy Conversion and Storage | Thermodynamics | Refrigeration and Air Conditioning | Electric Mobility

RESEARCH EXPERIENCE

Research Assistant, **Jul. 2023 to Present**
Sustainable Energy Engineering Research Group, University of Nigeria
Projects
Estimating the Potential for Maintaining Indoor Comfort in Smart Green Buildings with BIPV

- Conducted simulation-based research on energy-efficient building design using OpenStudio (HVAC load modelling) and the PVGIS tool (solar energy assessment).
- Modelled the energy requirements of a 2-storey, 11-room residential green building across 8 international climate zones (Nigeria, China, Egypt, USA, Mexico, India, Bulgaria, Germany).
- Designed and evaluated three HVAC operational strategies (continuous operation, living-space operation, smart occupancy-based operation), demonstrating up to 76.6% reductions in HVAC loads through energy-efficient scheduling.
- Integrated roof-mounted BIPV into the building model, assessing annual energy surpluses/deficits and the effect of roof slope and additional PV space on energy balance.
- Quantified the financial implications of BIPV integration, showing energy cost reductions of 10.2%-995.9%, with deficits converted to surpluses in most locations after smart control implementation.

Feasibility of Dual-Function BIPV for Indoor Thermal Comfort and Electric Vehicle Energy Demands in Nigerian Urban Centres

- Extended the previous study (as an undergraduate research assistant) by integrating EV charging analysis with BIPV energy surplus.
- Calculated monthly energy surpluses after HVAC demands and allocated them to EV charging (Tesla Model 3 LR AWD).
- Conducted state-based analysis (Abuja, Anambra, Lagos) using maximum travel distances (130–202 km).
- Performed a coordinate-based nationwide analysis, running 120+ location-specific simulations across Nigeria.
- Addressed missing/invalid data points (e.g., ocean or river coordinates) by applying Python-based interpolation and extrapolation.
- Produced bar charts and contour maps to represent spatial variations in EV charging potential and GHG savings using Pandas and Matplotlib.

Graduate Research Assistant, **Apr. 2025 to Present**
Circular Economy Powered Renewable Energy Centre (CEPREC)
Projects
Comparative Assessment of Battery and PCM Energy Storage in Off-grid PV-Powered Cold Rooms

- Developed a dynamic TRNSYS model of the Manamuz Electric PV-powered cold room in Enugu State, using the facility data.
- Compared the existing 52.5 kWh lithium-ion battery energy storage system with PCM thermal storage, selected to match the cold room's -5°C operating setpoint and suitable for envelope integration.
- Sized the PCM to deliver the same cooling energy as the battery system and simulated storage performance across product loads of 0–3000 kg.
- Evaluated refrigeration energy, PV utilisation, PV dumped energy, storage behaviour, temperature stability, compressor cycling, and system performance under varying product loads.
- Results showed that PCM storage reduced annual refrigeration energy consumption by up to 7.1% at 3,000 kg product loading, while providing smoother temperature control and reducing compressor cycling compared with battery storage.

- Battery storage achieved a shorter initial cooling period than PCM storage, demonstrating its advantage in rapidly bringing the cold room to the required operating temperature.
- Engaged with Manamuz Electric to recommend PCM thermal storage, with the company considering integration in future cold rooms.
- Initiated discussions with the industry partner on obtaining operational/IoT data from the deployed system to validate the TRNSYS model.

Decarbonising the Cold Chain: Battery and PCM Storage, System Autonomy, and EV Charging

- Extended the cold room energy storage study to evaluate system autonomy during solar outages and surplus PV utilisation for EV charging.
- Modelled battery and PCM storage under varying product loads and evaluated storage autonomy during periods of zero PV generation, alongside the potential for surplus PV energy to support electric produce transportation.
- Assessed EV charging potential using a Tesla Model 3 Long Range AWD to support low-carbon cold chain transportation.
- PCM storage provided longer system autonomy and greater EV charging potential than battery storage.

Hybrid Battery-PCM Energy Storage

- Developed and simulated hybrid battery-PCM energy storage configurations with varying battery-to-PCM contributions (10–90%).
- Evaluated the combined electrical and thermal storage performance under different operating conditions and product loads.
- Results demonstrated that hybrid storage addressed the prolonged initial cooling period observed with PCM-only storage while maintaining stable cold-room temperatures.

WORK EXPERIENCE

Graduate Assistant, **Jun. 2025 to May. 2026**

Nigerian Airspace Management Agency (NAMA), Chinua Achebe International Airport, Umueri, Anambra State (NYSC)

- Receives and verifies that the provided flight plans are free of errors, files them, and submits them to the control tower.
- Records and compiles daily statistics on the number of passengers, crew members, and flights using Excel.
- Prepares the monthly report for the Aeronautical Information Service (AIS), communications, and commercial departments.

Intern, **May. to Aug. 2023**

Scientific Equipment Development Institute, Enugu *[View Recommendation Letter]*

- Served as team lead for students undergoing industrial training, coordinating and supervising team members in carrying out day-to-day activities across assigned sections.
- Designed and assembled various machine components using SolidWorks in the design section.
- Manufactured machine parts, tools, and fixtures in the machine shop using lathes and drilling presses.
- Acquired fundamental knowledge of CNC and laser cutting machines in the advanced manufacturing section

Freelance Mathematics Expert **Jun. 2022 to Jan.2024**

Photomath (Remote)

- Solved high school and college-level math problems (elementary, algebra, statistics, geometry and calculus)
- Provided step-by-step solutions, ensuring clarity and accuracy for global student users.
- Strengthened ability to communicate technical concepts in simple terms.

TEACHING EXPERIENCE

Classroom Tutor (*Mathematics and Physics***)** **May. 2019 to Sep. 2021**

Saint Maria Goretti Secondary School, Agbalaenyi, Oji-River, Enugu State, Nigeria.

- Provided tutorials in Mathematics and Physics to over 150 students.
- Focused on targeted preparation for WAEC and JAMB examinations.
- Successfully guided 96% of the students to pass Mathematics and Physics, improving their chances of university admission.

LEADERSHIP AND VOLUNTEERING EXPERIENCE

Pioneer and Lead Team Member, **Jun. 2022 to Mar. 2024**

Academic Carryover Eradication Network (ACENetwork) *[View Leadership Award]*

- Co-founded ACENetwork, a peer-support initiative established to help students retaking courses overcome academic difficulties.
- Led tutorial sessions in Thermodynamics (ME 261, ME 361, ME 461 and ME 561) and other core Mechanical Engineering courses.
- Contributed to a significant reduction of over 90% in course carryovers among students supported by the initiative.
- Recognised by the Department of Mechanical Engineering with a leadership award for the initiative's contribution to student academic performance and departmental development.

Classroom Tutor (*Volunteer*), **Mar. 2021 to Feb. 2024**

Department of Mechanical Engineering, University of Nigeria, Nsukka

- Volunteered to teach and support my classmates across several core Mechanical Engineering courses, including Thermodynamics (ME 261, ME 361, ME 461, ME 561), Refrigeration and Air Conditioning (573), and Energy Conversion Principles (569).
- Organised group study sessions and one-on-one tutoring to clarify difficult concepts and enhance exam preparedness.

- Helped classmates achieve higher performance outcomes, building a culture of collaboration and academic excellence.

PUBLICATIONS, PEER-REVIEWED

- **Ngwu, N.**, Segnon, J.R., Nwokedi, M.E., Njoku, H. (2026). Feasibility of Dual-Function BIPV for Indoor Thermal Comfort and Electric Vehicle Energy Demands in Nigerian Urban Centres. In: Lagaros, N.D., Alrousan, R.Z., Georgantzinos, S.K. (eds) Proceedings of the International Conferences on Digital Technology Driven Engineering 2025. DiGiCES 2025. Lecture Notes in Civil Engineering, vol 877. Springer, Cham. https://doi.org/10.1007/978-3-032-28520-1_14
- **Ngwu, N.**, Ahanonu, M., Chen, K., Sang, Y., Njoku, H., & Pei, G. (2025, January 20). Estimating the Potentials for Indoor Comfort in Smart Green Buildings with BIPV. 10th Virtual International Conference on Science, Technology and Management in Energy (eNergetics 2024), Virtual. https://doi.org/10.5281/zenodo.14732925
- Eke, M., Umemba, U. I., Omeje, P. C., Odenyi, E. C., & **Ngwu, N. C** (2025). Investigation into the performance and emission of diesel engine operating at different blends of Egusi melon (Citrullus lanatus) biodiesel. *Intellectual Journal of Energy Harvesting and Storage*, *3*(1), 38-45. https://doi.org/10.11591/ehs.v3i1.pp38-45

PUBLICATIONS, UNDER REVIEW

- **Ngwu, N.**, Njoku, H., Oyinlola, M., Shire, S., Mbamalu, U., Aliozor, C. (2026). Comparative Assessment of Battery and PCM Energy Storage in Off-grid PV-Powered Cold Rooms. African International Conference for Clean Energy and Energy Storage.

CONFERENCE PRESENTATIONS

- ***"Comparative Assessment of Battery and PCM Energy Storage in Off-grid PV-Powered Cold Rooms"***: Oral Presentation, African International Conference for Clean Energy and Energy Storage (September 24, 2026) [View Certificate of Participation]
- ***"Feasibility of Dual-Function BIPV for Indoor Thermal Comfort and Electric Vehicle Energy Demands in Nigerian Urban Centres"***: Video Presentation, International Conference on Digital Technology Driven Engineering (December 18, 2025) [View Certificate of Participation]
- "***Estimating the Potentials for Indoor Comfort in Smart Green Buildings with BIPV***": Poster Presentation, 10th Virtual International Conference on Science, Technology and Management in Energy (November 25, 2024)

SCHOLARLY AFFILIATIONS

- Nigerian Society of Engineers.
- Society of Petroleum Engineers.
- Circular Economy Powered Renewable Energy Centre (CEPREC)
- Alumni Aspire Institute.

CERTIFICATIONS

- Scientific Writing, Cureus Journals (Part of Springer Nature) [View Certificate] **Sep. 2025**
- Renewable Energy and Green Building Entrepreneurship (Coursera), Duke University [View Certificate] **Aug. 2025**
- Aspire Leaders Program, Aspire Institute [View Certificate] **Aug. 2025**
- Python Programming Certification, National Institute of Information Technology (NIIT) [View Certificates] **Jul. 2021**

ACHIEVEMENTS/AWARDS

- Petroleum Technology Development Fund In-Country Scholarship Award for Masters [View Award Notification] **Oct. 2026**
- Intra-Africa – Mobility in Sustainable Energy & Materials, Energy, Policy, Climate Change, Energy Economics and Environment (SMECC2E) Scholarship Award for training in HOMER and LEAP Software [View Award Nomination Letter] **Sep. 2026**
- Circular Economy Powered Renewable Energy Centre (CEPREC) (2025) – Financial support for Master's Research Assistantship and Mentorship in Renewable Energy Systems. **Apr. 2025**
- Departmental Prize to the best graduating student in the Department of Mechanical Engineering [View Award] **Dec. 2024**
- Okenze Ezinne Innocent Nneji Chukwu Prize to the best graduating student in the Department of Mechanical Engineering [View Award] **Dec. 2024**
- HOD's Honours Award for the best-performing students in Mechanical Engineering [View Awards] **Mar. 2021, Apr. 2022, Apr.2023**

SKILLS

Energy system modelling | Building energy modelling | TRNSYS | OpenStudio | Python | Data analysis | Excel | Lathe operation | LaTeX | Academic writing | Problem-Solving skills | Analytical skills | Communication skills| Leadership | Teamwork